Hypothesis

i) Part B

I predict that the number of infected people will reach its peak around day 40~45, with about 450~500 people. And infection will increase initially, because there are many susceptible people. As more people recover and fewer susceptible people remain, the number of infected people will begin to decrease.

ii) Part D.

I would expect serval waves of intention, especially during the 1500 days simulation. This is because births continuously add new susceptible people to the population, allowing the disease to spread again after the 1st wave. It will also delays the infections, which may affect the timing of the waves.

iii) Part E

I think that β will have a greater effect on the infection peak than the recovery rate γ. A high β allows the disease to spread more quickly and infect more people at the same time. A high γ reduces the number of infected people by letting them recover faster.